8.3 Probability

The word 'probability' or 'chance' is very commonly used in day-to-day conversation and generally people have a rough idea about its meaning. For example, we come across statements like

"Probably it may rain tomorrow";

"The chances of teams A and B winning a certain match are equal";

All these terms – possible, probable, etc., convey the same sense, i.e., the event is not certain to take place or, in other words, there is uncertainty about happening of the event in question. In Layman's terminology the word 'Probability' thus can notes that there is uncertainty about what has happened. However, in mathematics and statistics we try to present conditions under which we can make sensible numerical statements about uncertainty and apply certain methods of calculating numerical values of probabilities.

Galileo (1564-1642), an Italian mathematician, was the first man to attempt quantitative measure of probability while dealing with some problems related to the theory of dice in gambling. The figure (8.1) given below represents the basic concepts of probability.

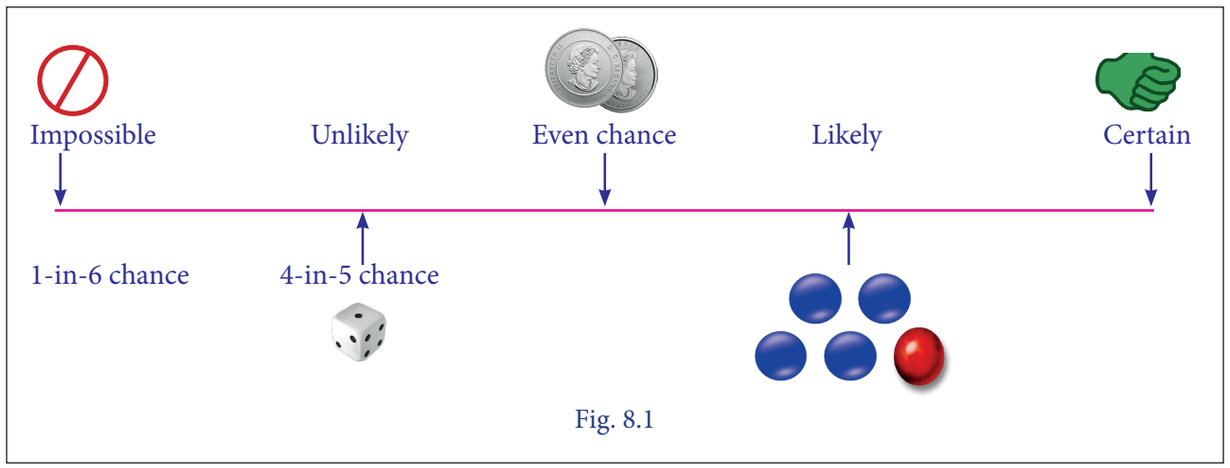


Fig. 8.1

8.3.1 Basic concepts of Probability

Recall

(i) Random Experiment

If an experiment or trial can be repeated under the same conditions, any number of times and it is possible to count the total number of outcomes, but individual result ie., individual outcome is not predictable, then the experiment is known as random experiment.

Example: Tossing a coin, throwing a die, selecting a card from a pack of playing cards, etc.

(ii) Outcome:

The result of a random experiment will be called an outcome.

(iii) Trial and Event:

Any particular performance of a random experiment is called a trial and outcome or combinations of outcomes are termed as events.

(iv) Exhaustive Events:

The total number of possible outcomes of a random experiment is known as the exhaustive events.

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